Muzaddidul Islam Chowdhury



I want to research electronic and energy materials and the devices built from them, from synthesis and characterisation to device modelling.

Open to funded PhD positions for 2027 and 2028 intakes. plabon.mic@gmail.com | linkedin.com/in/muzaddidul | scholar.google.com/citations?user=mdswSJ8AAAAJ

2018–19 **Lab** — $BiFeO_3$ nanoparticles for solar cells

2018–21 **Devices** — Carbon-nanotube FET modelling, 2 IEEE papers

2021–22 **Storage** — MXene battery-electrode review

2021–23 **Factory** — Process Analyst, production engineering at BAT

2023–25 **Supply network** — Scheduler for a USD 67M+ material portfolio at BAT

2025–now **AI for manufacturing** — Project Manager at Aivelon

My undergraduate research covered multiferroic $BiFeO_3$ nanoparticles for photovoltaics and non-ballistic carbon-nanotube field-effect transistors, and led to five publications in IEEE, IOP and JNEP venues, including a review of MXene heterostructures for battery electrodes. Since graduating I have worked in manufacturing at British American Tobacco and now at Aivelon, which taught me what a material or device needs to survive outside the lab: tight process control, reliable supply and data at every step.

5 publications: 1 journal article, 2 IOP proceedings, 2 IEEE conference papers.

Hands-on synthesis and characterisation: sol-gel, XRD, FESEM, TEM, EDS, UV-Vis-NIR.

Device modelling of CNT-FETs with high-κ dielectrics; COMSOL solar cell modelling.

What I want to research

1. **Bandgap and phase engineering of $BiFeO_3$ absorbers**

   Using doping, annealing schedules and nanocomposites to narrow the bandgap and remove secondary phases, then testing the materials in thin-film solar cell architectures.

2. **Electrostatic control in nanotube and 2D-channel transistors**

   High-κ gate dielectrics, off-state leakage and tunnelling in CNT and 2D-material FETs, extending my modelling work to fabricated devices.

3. **MXene and carbon nanocomposite electrodes for lithium-ion batteries**

   Heterostructured and biomass-derived carbon anodes that balance capacity, cycle life and sustainability, built on the MXene review.

Publications

MXene based heterostructures for electrode materials of batteries: a review
M. A. H. Shanto, **M. I. Chowdhury**, A. B. Antu, N. R. Niloy, N. Alam, M. A. Ullah, M. S. Anowar
IOP Conference Series: Materials Science and Engineering, vol. 1225, 012018 (IC-MAME 2021)
Open access
Journal-indexed proceedings, 2022 **DOI**

A study of OFF-state current and impact of dielectric constant on I_{ON}/I_{OFF} ratio of carbon nanotube field effect transistor
M. A. H. Shanto, **M. I. Chowdhury**, N. R. Niloy, N. Alam, M. J. Nayem, S.-A. Imam
2021 6th International Conference for Convergence in Technology (I2CT), Pune, India. IEEE
IEEE conference, 2021 **DOI** **PDF**

Multiferroic bismuth ferrite nanocomposites as a potential photovoltaic material
N. R. Niloy, **M. I. Chowdhury**, M. A. H. Shanto, J. Islam, M. M. Rhaman
IOP Conference Series: Materials Science and Engineering, vol. 1091, 012049 (ICIRMCT 2021)
Journal-indexed proceedings, 2021 DOI PDF

Structural and optical characterization of multiferroic $BiFeO_3$ nanoparticles synthesized at different annealing temperatures
N. R. Niloy, **M. I. Chowdhury**, S. Anowar, J. Islam, M. M. Rhaman
Journal of Nano- and Electronic Physics, vol. 12, no. 5, 05015 (2020)
Journal article, 2020 DOI PDF

Effect of channel length and dielectric constant on carbon nanotube FET to evaluate the device performance
M. A. H. Shanto, N. R. Niloy, R. Islam, **M. I. Chowdhury**, M. M. Abrar, S.-A. Imam
2020 4th International Conference on Electronics, Materials Engineering & Nano-Technology (IEMENTech), Kolkata, India, pp. 1–4. IEEE
IEEE conference, 2020 DOI PDF

Research experience

Properties of multiferroic bismuth ferrite nanoparticles and their prospect in photovoltaic applications Jul 2018 – Aug 2019
Undergraduate thesis, AUST. Supervisor: Dr. Md. Meganur Rhaman, Professor, EEE
Synthesised $BiFeO_3$ nanoparticles by the sol-gel method and annealed them at 500, 600 and 700 °C. Characterised structure with XRD, FESEM, TEM and EDS, and optical properties with UV-Vis-NIR. The 500 °C samples showed fewer secondary phases and a favourable bandgap for absorber layers. Led to the JNEP 2020 and IOP 2021 papers.

Improving the performance of single-wall carbon nanotube FETs Mar 2018 – Jun 2019
Undergraduate research, AUST. Supervisor: Safayat-Al Imam, Assistant Professor, EEE
Modelled non-ballistic effects (scattering, strain and tunnelling) in CNT-FETs. Showed how high-κ gate dielectrics strengthen electrostatic control, shorten the usable channel and suppress off-state current. Led to the IEMENTech 2020 and I2CT 2021 papers.

Engineering projects

- **1D model of a silicon solar cell in COMSOL Multiphysics 5.5.** Jan – Mar 2019, supervised by Safayat-Al Imam. Slides Report
- **Dual-axis sun-tracking solar panel with light sensors and stepper motors.** Dec 2018 – May 2019, supervised by Nahyan Al Mahmud. Video
- **Arduino-controlled irrigation using soil-moisture sensing.** May – Sep 2018, supervised by Md. Tashfiq Bin Kashem. Slides Report
- **16:1 multiplexer in transmission-gate logic, schematic and layout in Cadence Virtuoso.** Coursework.

Industry experience

Project Manager Nov 2025 – present
Aivelon, Dhaka
Software company building SaaS and generative-AI-as-a-service products, currently focused on manufacturing clients.

- Lead the delivery team and client relationships, turning manufacturing clients' operational problems into software requirements and delivery plans.
- Supervise an engineering intern and work through the backend architecture (Java, Spring Boot) with him.

Wrapping Material Scheduler, Supply Network Operations Mar 2023 – Oct 2025
British American Tobacco Bangladesh, Dhaka

Planned packaging materials for a plant producing about 60 billion sticks a year.

- Built and ran inventory models for a USD 67M+ material portfolio, reaching 85% forecast accuracy from demand and consumption data.
- Coordinated 10+ foreign suppliers across Asia and Europe, from import documents and letters of credit to port clearance.

Process Analyst, Production Engineering Jun 2021 – Feb 2023
British American Tobacco Bangladesh, Dhaka

Spare parts and maintenance readiness for the production line.

- Reduced overseas OEM spare-parts lead time from 88 to 71 days, improving maintenance readiness by 18%.
- Automated material requirement planning in Excel VBA, reaching 100% procurement accuracy.

Maintenance Engineer Oct 2019 – Jul 2021
Apsara Trading Ltd., Dhaka

Education

B.Sc. in Electrical and Electronic Engineering 2015 – 2021
Ahsanullah University of Science and Technology (AUST), Dhaka

Major in electronics. Thesis on bismuth ferrite nanoparticles for photovoltaics.

Skills

Synthesis and characterisation

Sol-gel synthesis, annealing; XRD analysis with FullProf and VESTA; FESEM, TEM, EDS; UV-Vis-NIR bandgap analysis

Simulation and design

COMSOL Multiphysics, Cadence Virtuoso, PSpice, Proteus, Simulink

Data and programming

Python, MATLAB, C++, OriginPro, Power BI